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EDUCATION

2026 – 2030
UC Berkeley

2022 – 2026
Palo Alto High School
• GPA: 4.0 / 4.0

SKILLS

Programming

• Linux CLI • ElectronJS
• Python • ThreeJS
• Numpy • Redux
• SciPy • React
• Tkinter • RTKQ
• PyGame • Tailwind
• JS • Framer
• TS • R3F
• HTML • Java
• CSS • Gradle
• NodeJS • GLSL
• OpenAI SDK
• Anthropic SDK
• Gemini SDK

Physics

• Newtonian (Calculus)
• Lagrangian

General Biology

General Chemistry

Graphic Design

Jeffrey Fan.

I make apps, art, robots, and more.

An aspiring creator looking to build things with the fundamentals of biology, chemistry, physics, and computers. Ready to make more tools for more people.

WORK EXPERIENCE

➤ Lawrence Berkeley National Laboratory

Feb 2024 – Present

Paid Student Research Assistant

- Operated on a large scale with supercomputing to evaluate quantum-based chemistry geometric optimization and frequency analysis of molecules to determine reaction efficiency and properties
- Worked on a succinct [CL](#) tool for batch submitting and tracking repetitive tasks to [Slurm](#)
- Performed statistical analysis of DNA structures using [alternate coordinate representations](#) to determine their [forms](#)
- Created generalized 2D reaction coordinate/trajectory generator by large data processing and automation, hosted on [website](#) for ease of access
- Optimized PETase enzyme via local/small-scale reaction simulation, proposed novel enzyme stability optimization parameter, working on paper independently

➤ EmdoorVR

Jul – Aug 2026

Intern/External Collaborator

- Worked on client/phone-side optimizations for communication with peripheral next-gen multimodal AI smart glasses
- Proposed and designed improvements to Android SDK, from spec to software
- Created custom Bluetooth/BLE and WiFi P2P integration protocols for device identification and local communication
- Designed educational curriculum for new company division

➤ A0 Systems – Hopper Division

Jun – Aug 2025

Paid Intern

- Worked on a small team developing a next-generation coding tool and project manager, prioritizing developer support from idea to product
- Worked on front-end with React, Redux, and Tailwind
- Worked on back-end ElectronJS with AI frameworks for generalized agent setup (multi-company SDK support: OpenAI, Anthropic, Gemini)
- Refactored and improved the robustness of the entire UI framework, fixed countless bugs and inefficiencies
- Created numerous features for the application entirely solo:
 - Gemini video analysis for bug discovery and description
 - Agentic-based idea → feature → component → code generation
 - Chatbox agents for project modification
 - Stage linking graphical feature to see traces in development history

AWARDS

- FRC Autonomous Award
- FRC Innovation in Control Award
- USACO Silver
- AIME Qualifier
- Congressional App Challenge Honorable
- FIC Honorable
- SVTT Honorable
- Paly Science Award
 - Biology H
 - Chemistry H
 - AP Physics C
- Congressional Art Competition Honorable
- Scholastic Art Gold Key
- Scholastic Art Silver Key
- PVSA
- National Merit Semifinalist
- National Merit Finalist Scholarship
- AP Scholar With Distinction

LANGUAGES

English – Fluent
Spanish – AP
Chinese – Fluent

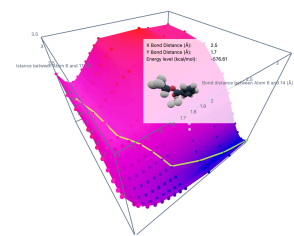
LEADERSHIP

- Software Captain @ FRC
- Group Lead @ COSMOS
- CTO and Group Lead @ Noteblast, ACF
- Lab Group Lead @ SSSIP

PROJECTS

➤ Reaction TrajFinder

Reaction TrajFinder is an ORCA output file post-processor that extracts energy data to produce a reaction pathway of least activation energy between two points. It works only on 2D bond distance scans. It is not meant to replace conventional MD simulations, but serves to add on to ORCA's existing geometric optimizer.



➤ LuxCM5

LuxCM5 is the custom software for custom-PCB cameras designed for FRC AprilTag detection and localization. Features UMich AprilTag detector optimizations and full frontend application. Connects over Ethernet with NetworkTables protocol for robot communication. Beats open-source solutions by 8x FPS and 4x latency, hitting up to 120 FPS and 20ms latency at high (2MP) resolution.



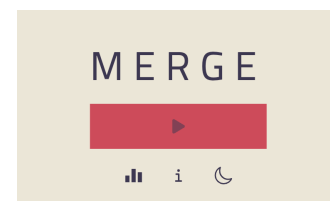
➤ Visual Art Gallery V2

The second iteration of my virtual art gallery, with a strong focus on light, immersion, and camera-like effects. Features a full 3D gallery with a tutorial for navigation, organization, and non-Euclidean space-looping effects! Check it out!



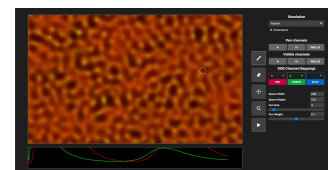
➤ Merge Game

A fun 2048-like Chrome extension game about merging shapes to create higher tiers. A good offline way to enjoy some downtime. Boasting over 3.6K installations in total and nearly 900 weekly users.



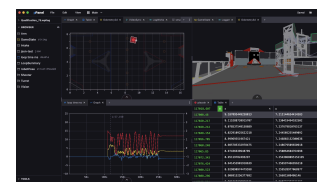
➤ Biotech Sandboxes

A sandbox environment where I simulated some mathematical models of biological systems. Notable models include the Gierer-Meinhardt model for animal coat pattern formation and the FitzHugh-Nagumo model for neuron action potential and heart tissue simulation.



➤ PeninsulaPortal

A complete FRC robot diagnostic and debugger, path generator, and competition pit display. Sleek, navigable design focused on user experience. Full end-to-end development from back-end DigitalOcean servers to front-end ElectronJS applications.



➤ FIRST Robotics Competition

2023 – 2026

Software Captain of 6036

- Management of a team of over 30 software recruits
- Robot competes in 3 annual regionals
- 2026: Top 100 of 3700 teams in the world, top 10 of 290 in California.
- Created a robust robot development environment
 - Necessary debugging and tuning tools built in for rapid testing in the short 2-month season
 - Modularity and collaboration-focused software architecture promotes sustainability
 - Versatile custom library speeds up development tenfold
- Created a front-end application for diagnostic interaction with the robot during a competition
- Developed an entire [scouting system](#) for information gathering during FIRST competitions
- Created a [fully custom vision system software](#) interacting with custom hardware for x8 performance improvement of AprilTag detection compared to open source solutions

➤ COSMOS UC Davis – Computational Biology

July 2024

Group Lead

- Led a group of 4 through modeling numerous biological systems
- Created a [front-end application](#) capable of simulating numerous mathematical models
 - Fitz-Nagumo neuron and heart action potential model
 - Gierer-Meinhardt animal coat pattern formation model
 - Boid and Viseck flocking model
- Presented an analysis on Gierer-Meinhardt pattern amplification frequency related to initial empirical model parameters to professors
- Wrote a [write-up](#) of our discoveries and learning process

➤ Applied Computing Foundation (ACF) – NoteBlast

2022 – 2024

CTO and Group Lead

- Solely created a [full-stack application](#) called Noteblast, providing musicians with free instrument tuning and built-in sightreading gameplay
- [Congressional App Challenge](#) Honorable mention
- [SVTTI](#) 9th Summit participant
- [FIC](#) Honorable mention
- Author of a [published paper](#) about music and education sustainability in JSR

➤ Stanford Summer Science Internship Program (SSSIP)

July 2023

Lab Group Lead

- Led a lab group of 4 through experiments and data gathering
- Organized and designed the final [presentation](#) for Stanford professors
- Developed an image processing algorithm (and later an [app](#)) to count transfected cells, which surprised camp mentors